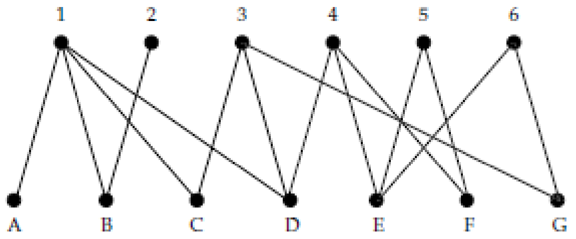
DSC– Machine Learning Data Mining
Fiche Social mining

Exercice 2

Given a two mode graph G2 = (⊥,T, E) où ⊥={1, 2, 3, 4, 5, 6} et T = {A, B, C, D, E, F, G}



Give the projection of G2 on ⊥ by adjacency list?

Exercice 3

Given a graph G = (E, V) with 11 vertices defined by its adjacency matrix :

	x1	**x2**	**x3**	**x4**	**x5**	**x6**	**x7**	**x8**	**x9**	**x10**	**x11**
x1	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
x2	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
x3	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
x4	**1**	**1**	**1**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**
x5	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**1**	**0**	**0**
x6	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**1**	**0**	**0**	**0**
x7	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**1**	**0**	**0**	**0**
x8	**0**	**0**	**0**	**0**	**0**	**1**	**1**	**0**	**0**	**0**	**0**
x9	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**1**	**1**
x10	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**
x11	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**
V(x)	**1**	**1**	**1**	**4**	**3**	**3**	**2**	**2**	**3**	**1**	**1**

1- Give the density and the diameter of G

2 - Give the degree distribution for this network

3- Give the local clustering coefficient of the vertices and the network average clustering coefficient of G

4- Give a definition of the degree centrality Give the node with the highest degree centrality

5- Give a definition of the betweeness centrality .Give the node with the highest betweeness centrality

Exercice 4 (E2015-16)

A social network having 9 vertices is represented by its adjacency matrix:

	1	2	3	4	5	6	7	8	9
1	0	1	1	1	0	0	0	0	0
2	1	0	1	0	0	0	0	0	0
3	1	1	0	1	0	0	0	0	0
4	1	0	1	0	1	0	0	0	0
5	0	0	0	1	0	1	1	1	0
6	0	0	0	0	1	0	1	1	0
7	0	0	0	0	1	1	0	1	1
8	0	0	0	0	1	1	1	0	1
9	0	0	0	0	0	0	1	1	0

1- Give the diameter for this network

2- Using the min Cut method, propose a partition of the network into two groups.
What is the well known drawback of this method? Give an example to illustrate this drawback. Which extensions have been introduced to overcome this drawback?

3- Apply the clique percolation method (CPM) with the parameter set to 3 to determine another community structure P for this network. What is the advantage of CPM? Give an example to illustrate this advantage.

4- In fact, we know that there are 2 communities in this network and this ground truth partition is defined by P* = {{1, 2, 3 },{4, 5, 6, 7, 8, 9}}
Compute the accuracy of P according to the pairwise community membership.